

Today's Agenda

- Transmission Line Impedance
- E & M waves
- Reflection Transmission

Review Impedance

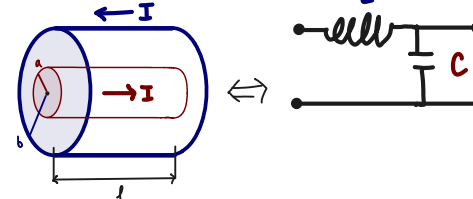
$$\frac{1}{Z_R} = R \quad \frac{1}{Z_C} = i\omega C \quad \frac{1}{Z_L} = i\omega L$$

$$\tilde{V} = \tilde{I} Z \quad \left\{ \begin{array}{l} V(t) = \text{Re}\{\tilde{V}\} \\ I(t) = \text{Re}\{\tilde{I}\} \end{array} \right.$$

Z 's in series add: $Z_{eq} = Z_1 + Z_2$

Z 's in parallel: $\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2}$

coaxial cable



recall,

$$C = \frac{2\pi\epsilon_0}{\ln(b/a)}$$

$$C_0 = \frac{C}{l} = \frac{2\pi\epsilon_0}{\ln(b/a)}$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)$$

$$L_0 = \frac{L}{l} = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)$$

Loop Rule:

$$V_{n-1} - V_n - V_L = 0$$

$$\hookrightarrow V_{n-1} - V_n = V_L = L_0 \frac{dI_n}{dt}$$

$$\frac{V_{n-1} - V_n}{dx} = L_0 \frac{dI_n}{dt}$$

$$\frac{\partial V}{\partial x} = L_0 \frac{\partial I}{\partial t}, \quad \frac{\partial^2 V}{\partial x^2} = L_0 \frac{\partial^2 I}{\partial t^2}$$

Junction Rule:

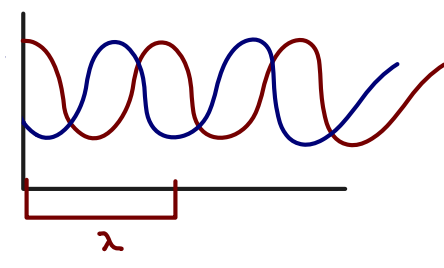
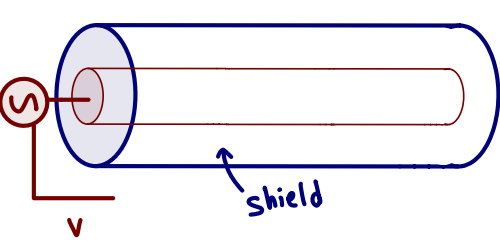
$$I_n = I_{n+1} + I_L$$

$$\hookrightarrow I_n = I_{n+1} + C_0 dx \frac{dV_n}{dt}$$

$$\frac{I_n - I_{n+1}}{dx} = C_0 \frac{dV_n}{dt}$$

$$\frac{\partial I}{\partial x} = C_0 \frac{\partial V}{\partial t}, \quad \frac{\partial^2 I}{\partial x^2} = C_0 \frac{\partial^2 V}{\partial t^2}$$

Transmission Lines



traveling waves
 $V = V_0 \cos(kx - \omega t)$

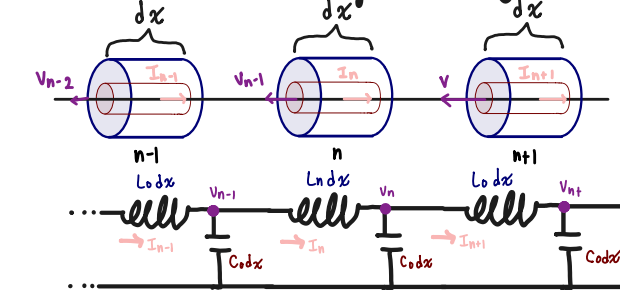
$k = \text{wave number} = \frac{2\pi}{\lambda}$

$\omega = \text{angular frequency} = 2\pi f$

speed = $c = \frac{\omega}{k} = \lambda f$

* $c = 3 \times 10^8 \text{ m/s} = 1 \text{ ft/nanosecond}$

consider cables as many small segments



solutions: Traveling Waves

$$V(x,t) = V_0 \cos(kx - \omega t) \text{ or } \tilde{V} = V_0 e^{i(kx - \omega t)}$$

$$\frac{\partial^2 \tilde{V}}{\partial x^2} = V_0 (ik)^2 e^{i(kx - \omega t)} = -k^2 \tilde{V}(x,t)$$

$$\frac{\partial^2 \tilde{V}}{\partial t^2} = V_0 (i\omega)^2 e^{i(kx - \omega t)} = -\omega^2 \tilde{V}(x,t)$$

$$\hookrightarrow \frac{1}{L_0 C_0} (-k^2 \tilde{V}) = (-\omega^2 \tilde{V}) \Rightarrow \frac{1}{L_0 C_0} \left(\frac{\omega}{k}\right)^2 = \omega^2$$

$$\hookrightarrow \frac{1}{L_0 C_0} = \left(\frac{\omega}{k}\right)^2 = c^2$$

$$\hookrightarrow c = \frac{1}{\sqrt{L_0 C_0}}$$

$$\tilde{V}_{n-1} - \tilde{V}_n = I_n Z_L$$

$$\tilde{V}_{n-1} - \tilde{V}_n = i\omega L_0 \tilde{I}_n$$

$$\hookrightarrow \frac{\partial \tilde{V}}{\partial x} = i\omega L_0 \tilde{I}$$

$$\tilde{V}(x,t) = V_0 e^{i(kx - \omega t)}$$

$$\tilde{I}(x,t) = \frac{V_0}{i\omega L_0} \frac{\partial}{\partial x} (e^{i(kx - \omega t)})$$

$$= \frac{V_0 ik}{i\omega L_0} e^{i(kx - \omega t)} = \frac{V_0}{\omega L_0} e^{i(kx - \omega t)}$$

$$\Rightarrow \frac{\tilde{V}(x,t)}{\tilde{I}(x,t)} = \frac{\omega}{k} \frac{1}{L_0} = \frac{1}{\sqrt{L_0 C_0}} \frac{1}{L_0} = \sqrt{\frac{L_0}{C_0}}$$

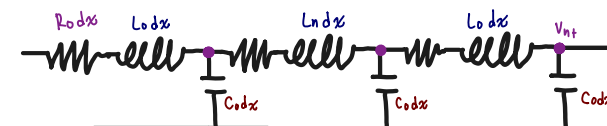
"characteristic Impedance"

* a long cable looks like a resistive load w/ $R = Z_0$

$$Z_0 = \sqrt{\frac{L_0}{C_0}} = \sqrt{\frac{\mu_0}{\epsilon_0} \ln\left(\frac{b}{a}\right) \frac{1}{2\pi}} = 50 \Omega$$

for $b \approx 2.3a$

for cable TV: $Z_0 = 75 \Omega$

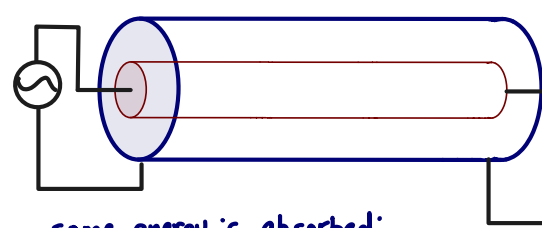


$R_0 = \text{Resistance unit length}$

- Resistance from metal
- energy absorbed by dielectric
- energy radiated away

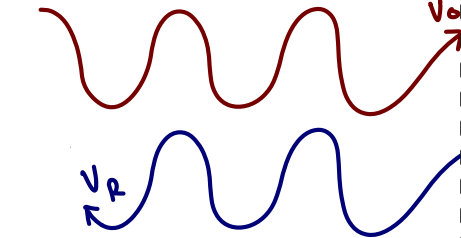
$$V(x,t) = V_0 e^{-\alpha x} \cos(kx - \omega t), \quad \frac{1}{\alpha} = \text{attenuation length}$$

The End of the Line



some energy is absorbed:
 $b \cdot Z_L$

Some is reflected back:



$$\tilde{V}_0 = V_0 e^{i(kx - \omega t)}$$

$$\tilde{I}_0 = \frac{1}{i\omega L_0} \frac{d\tilde{V}_0}{dx} = \frac{\tilde{V}_0}{Z_0}$$

$$\tilde{V}_L = \tilde{V}_0(x=0,t) + \tilde{V}_R(x=0,t)$$

$$\tilde{I}_L = \tilde{I}_0(x=0,t) + \tilde{I}_R(x=0,t)$$

$$\tilde{V}_R = V_R e^{i(kx - \omega t)}$$

$$\tilde{I}_R = \frac{1}{i\omega L_0} \frac{d\tilde{V}_R}{dx} = \frac{V_R (-ik)}{i\omega L_0} e^{i(kx - \omega t)} = -\frac{\tilde{V}_R}{Z_0}$$

$$\tilde{V}_{tot} = \tilde{V}_0(x,t) + \tilde{V}_R(x,t)$$

$$\tilde{I}_{tot} = \tilde{I}_0(x,t) + \tilde{I}_R(x,t)$$

$$\tilde{V}_L = \tilde{V}_0(x=0,t) + \tilde{V}_R(x=0,t)$$

$$\tilde{I}_L = \tilde{I}_0(x=0,t) + \tilde{I}_R(x=0,t)$$

$$\rightarrow \text{for a Load Impedance } Z_L$$

$$\text{must be that } Z_L = \frac{\tilde{V}_L}{\tilde{I}_L} = \frac{\tilde{V}_0 + \tilde{V}_R}{\tilde{I}_0 + \tilde{I}_R} = \frac{\tilde{V}_0 + \tilde{V}_R}{\frac{\tilde{V}_0}{Z_0} - \frac{\tilde{V}_R}{Z_0}} = \frac{\tilde{V}_0 + \tilde{V}_R}{\tilde{V}_0 - \tilde{V}_R} Z_0$$

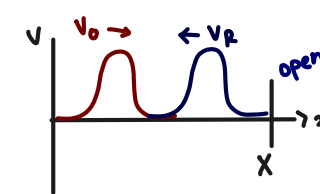
$$\Rightarrow \frac{\tilde{V}_R}{\tilde{V}_0} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\frac{\tilde{V}_R}{\tilde{V}_0} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

examples:

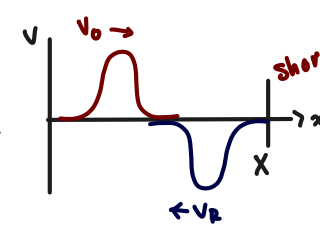
open-circuit ($Z_L \rightarrow \infty$)

$$\frac{\tilde{V}_R}{\tilde{V}_0} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L}{Z_L} = 1$$



short-circuit ($Z_L \rightarrow 0$)

$$\frac{\tilde{V}_R}{\tilde{V}_0} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{-Z_0}{Z_0} = -1$$

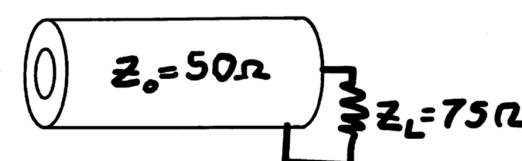


PHYS0525

Transmission Lines Practice Problems

Problem #1

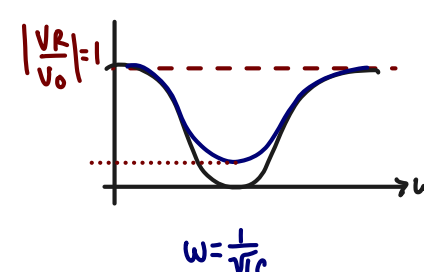
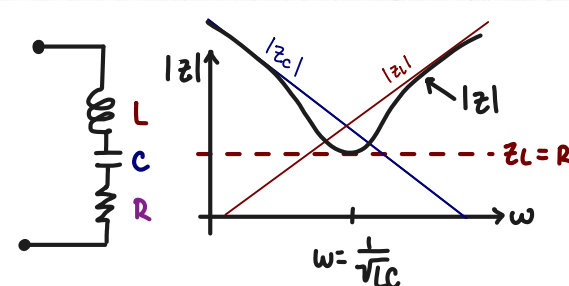
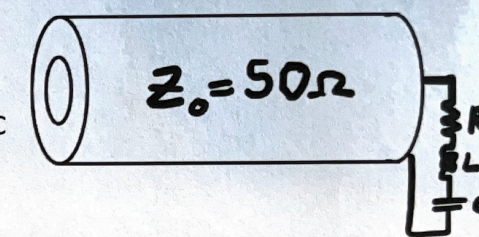
A 1 volt amplitude wave is traveling down a coaxial cable with characteristic impedance of $Z_0 = 50 \Omega$. The cable is terminated in a load $Z_L = 75 \Omega$. What is the amplitude of the reflected wave?



$$1) \frac{\tilde{V}_R}{\tilde{V}_0} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{75 - 50}{75 + 50} = \frac{25}{125} = \frac{1}{5}$$

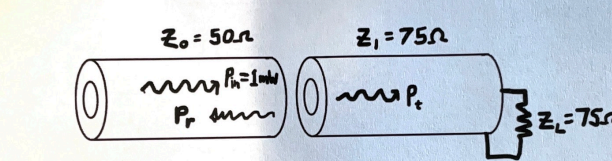
Problem #2

A 1 volt amplitude wave is traveling down a coaxial cable with characteristic impedance of $Z_0 = 50 \Omega$. The cable is terminated in a load Z_L consisting of an RLC series circuit with $R = 50 \Omega$, $L = 1 \text{ mH}$, and $C = 1 \text{ nF}$. Sketch the amplitude of the reflected wave as function of frequency.



Problem #3

A wave carrying a power of 1mW is traveling down a 50 Ohm coaxial cable. This cable is connected to second 75 Ohm coaxial cable, which is terminated in a 75 Ohm load. How much power is dissipated by the load resistor? What is the RMS voltage across the load?



$$\text{via problem } \frac{\tilde{V}_R}{\tilde{V}_0} = 0.2, \quad P_{in} = \frac{\tilde{V}_0^2}{Z_0}$$

$$P_{trans} = P_{in} - P_{ref} = 0.96 \text{ mW}, \quad \text{so: } \langle P_{load} \rangle = \frac{V_{RMS}^2}{Z_L} \Rightarrow V_{load} = \sqrt{Z_L \langle P_{trans} \rangle} = \sqrt{(0.96 \times 10^{-3}) 75} = \sqrt{0.072} \approx 0.268 \text{ V}$$

* Because the load is a 75 Ohm AND the 2ND line is 75 Ohm, there is no reflection @ the load